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Digital Electronics & VLSI
Codec Technologies India
Internship 2026

FPGA Traffic-Light Controller with Emergency Priority

Verilog FSM, debounced
override, iCE40-HX8K place &
route

AT A GLANCE

39 / 39 pass

Testbench checks (Icarus)

0

Safety assertion fires

318 / 7680

iCE40 logic cells (nextpnr)

109.42 MHz

iCE40 Fmax (nextpnr)

12.37 ns

icetime clk->pad path

109 / 128

Xilinx LUT / FF (yosys)

11

FSM states

10 ms, 2-FF sync

Debounce

VERDICT

SIM PASS

P&R OK

TIMING MET

DECLARED SUBSTITUTION

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1. Objective and brief mapping

The brief asked for a 4-way intersection controller in Verilog with a pedestrian phase, parameterised timings, synchronous reset, a debounced and synchronised emergency-vehicle override per approach that hands green to the emergency approach only through yellow and all-red, a 1 Hz tick from a 12 MHz iCE40 clock, a 7-segment countdown, a testbench covering normal cycle, override mid-green, override during yellow and two simultaneous emergencies with a no-conflicting-green assertion, and synthesis for both iCE40 (real place and route) and a Spartan-style LUT/FF count.

I implemented all of it. Everything that produced a number in this report was run in the build container with Icarus Verilog 14, yosys 0.68, nextpnr-ice40 and icetime. Vivado and ModelSim artefacts are supplied but were not run (section 6).

2. Architecture

The FSM is a single always-block Moore machine with binary-encoded states and a 6-bit phase timer that is reloaded on every state change and decremented on each 1 Hz tick. Lamp outputs are registered from the next-state vector, so a lamp never glitches and always agrees with the state. Emergency requests are reduced to two axis requests: want_ns = N or S (highest priority) and want_ew = (E or W) and not want_ns, which implements the N > S > E > W order with two gates.

```
emg_in[3:0] -> sync_debounce x4 -> emg_db ---+ +-----+> ns_light {R,Y,G}
(N,S,E,W)    2 FF + 120000-cycle count +----->| traffic_fsm |> ew_light {R,Y,G}
12 MHz clk   -> clk_div_tick (1 Hz pulse) ----->| 11 states |> ped_walk / ped_flash
rst_btn_n    -> 2 FF sync -> synchronous rst ----->| |> remaining[5:0] -> seg7 -
+-----+> emg_active, emg_idx
```

```
ring: NS_G(0) -> NS_Y(1) -> AR1(2) -> EW_G(3) -> EW_Y(4) -> AR2(5) -> WALK(6) -> FLASH(7) -> AR3
hold: EMG_NS(9) entered from NS_G / AR1 / AR2 / AR3 ; EMG_EW(10) entered from EW_G / AR1 / AR2 /
```

Request arrives during	Action taken	Why it is safe
Green on requested axis	Hold green in EMG_NS / EMG_EW, display E	No lamp change at all
Green on the other axis	Cut green, full yellow, all-red, then emergency green	Yellow and all-red are never shortened
Yellow	Yellow runs to completion, then all-red, then emergency green	Drivers already committed to stopping
Pedestrian WALK	Jump to don't-walk flash clearance, all-red, emergency green	Pedestrians get the clearance interval
Two requests together	Higher priority first; the other is served after yellow + all-red	Requests are latched by the debouncers
Request released	Emergency green ends via yellow + all-red; ring resumes	Same exit path as a normal phase

3. Implementation details

- sync_debounce: two flip-flops contain metastability on the asynchronous transponder line; the output flips only after STABLE_CYCLES (120 000 = 10 ms at 12 MHz) consecutive agreeing samples. Counter width is derived from the parameter with a constant clog2 function so the same source serves 12 MHz (iCE40) and 100 MHz (Basys3).
- clk_div_tick: 24-bit counter producing a one-cycle pulse every TICK_DIV clocks. A pulse rather than a divided clock keeps the whole design in one clock domain, which is what lets nextpnr report a single Fmax.
- seg7_countdown: 0..63 seconds are split into two BCD digits with a six-compare chain (no divider), then a shared hex-to-segment function. In emergency hold the digits read E0..E3 for N, S, E, W.
- All resets are synchronous; the reset button is itself synchronised through two flip-flops before use.
- Phase lengths, tick divider and debounce length are module parameters, overridden by the testbench (1 tick = 10 clocks, debounce = 4 clocks) so a full scenario set runs in 105 us of simulated time and a 73 kB VCD.

4. Verification

tb_traffic_top.v drives the board-level top, measures the number of ticks spent in every phase and compares it with the parameter, and carries three always-on assertions: NS and EW are never both non-red, WALK is never on while a vehicle lamp is green or yellow, and each lamp group is one-hot. Any violation calls \$fatal, so run_all.sh cannot report PASS with a conflict present. Result: 6 scenarios, 39 checks, 0 errors.

Scenario	What is checked	Result
T1 normal cycle	phase lengths 2,8,3,2,8,3,2,5,2,2 ticks; tens digit 0	pass
T5 glitch	2-clock pulse on W does not reach the FSM	pass
T2 E mid NS-green	NS_Y 3 ticks, AR 2, EW held green 6+ ticks, display E2, resume via EW_Y/AR to WALK	pass
T3 N during NS yellow	yellow completes, AR, EMG_NS green; release -> NS_Y, AR, EW_G	pass
T4 N + E during EW green	EW_Y, AR, North first (emg_idx 0); N released -> NS_Y, AR, East (idx 2); ring resumes	pass

KEY TAKEAWAY Every axis change passes through yellow + all-red by construction; the override never produced a conflicting green in 39 checks and the design runs at 109 MHz on an HX8K (12 MHz needed).

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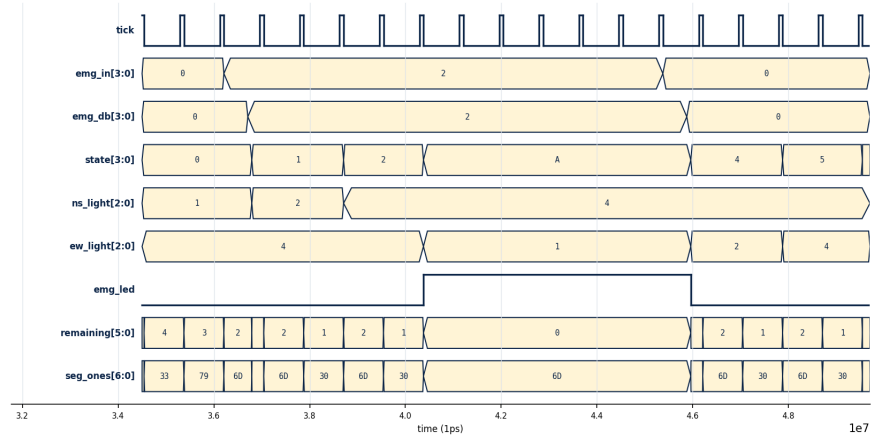
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Scenario	What is checked	Result
T6 S during WALK	WALK cut to FLASH (2 ticks), AR, South served	pass

T2: East emergency mid NS-green -> NS yellow -> all-red -> EW held green (state A) -> release -> yellow/all-red -> PED



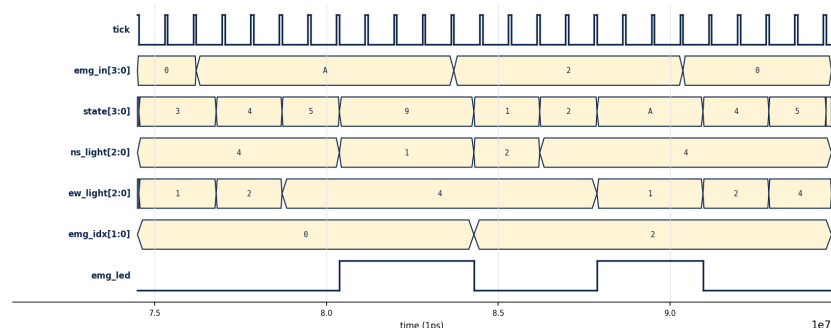
T2: East request during NS green. state 0 -> 1 (yellow, 3 ticks) -> 2 (all-red) -> A (EW emergency hold, remaining = 0, digit shows '2') -> 4 -> 5 after release.

5. Synthesis and implementation results

The two iCE40 figures differ because nextpnr reports the register-to-register clock-domain Fmax while icetime -mtr also includes the clock-to-output-pad path through the 7-segment decoder. Both are two orders of magnitude above the 12 MHz the board supplies. Flip-flop count is dominated by the 24-bit tick divider (24 FF) and four 17-bit debounce counters (68 FF plus 8 synchroniser flops); the controller itself needs 4 state bits, a 6-bit timer, 2 index bits and 8 lamp registers.

Target / tool	Resource result	Timing result
iCE40-HX8K, yosys synth_ice40	258 SB_LUT4, 128 DFF, 107 SB_CARRY	-
iCE40-HX8K CT256, nextpnr-ice40 with real .pcf	318 / 7680 logic cells (4 %), 30 / 206 IO, 1 global buffer	Fmax 109.42 MHz, 12 MHz constraint PASS
iCE40-HX8K, icetime -mtr	13 logic levels on longest path	12.37 ns clk -> seg_ones[1] pad (80.82 MHz incl. output pad)
Xilinx xc7 (Basys3 class), yosys synth_xilinx	109 LUT (31 LUT6, 26 LUT5, 31 LUT4, 11 LUT3, 10 LUT2), 128 FF, 28 CARRY4	not placed (no Vivado here)

T4: N+E simultaneous during EW green: EW_Y -> AR -> NS served (idx 0) -> N released -> NS_Y -> AR -> E served (idx 2)



T4: N and E requests arrive together during EW green (state 3). EW yellow (4) and all-red (5) run in full, North is served first (state 9, emg_idx 0); when N releases, NS yellow (1) and all-red (2) precede the East hold (state A, emg_idx 2).

6. Declared substitutions and how to run on the target tools

- Vivado / Basys3: not installed here. Run vivado -mode batch -source vivado/create_project.tcl; it adds rtl/*.v, vivado/basys3.xdc (100 MHz W5 clock, buttons U18/T18/U17/T17/W19, LEDs, shared segment bus), sets TICK_DIV = 100 000 000 and DEB_CYCLES = 1 000 000 as generics and writes utilization and timing reports. The Xilinx LUT/FF numbers above are from yosys synth_xilinx, not Vivado.
- ModelSim / Questa: not installed here. vsim -c -do modelsim/run.do compiles the same sources and testbench. All simulation results in this report are from Icarus Verilog 14.
- No physical board: the iCE40 figures come from nextpnr/icetime on the real HX8K-CT256 device database; icepack was run to prove the bitstream builds, but the .bin is not kept in the repo.

7. Learnings and next steps

Registering the lamp vector from the next-state function was the cleanest way to get glitch-free outputs and it also removed a bus-lane artefact in the waveform tool. Deriving counter widths from parameters with a constant function avoided the classic mistake of a 17-bit debounce counter silently overflowing when moved to a 100 MHz board. Next steps: time-multiplex the two digits on the Basys3 shared segment bus, add a vehicle-detector input to skip the EW green when empty, and log override events over UART for a traffic-department audit trail.

KEY TAKEAWAY Every axis change passes through yellow + all-red by construction; the override never produced a conflicting green in 39 checks and the design runs at 109 MHz on an HX8K (12 MHz needed).